

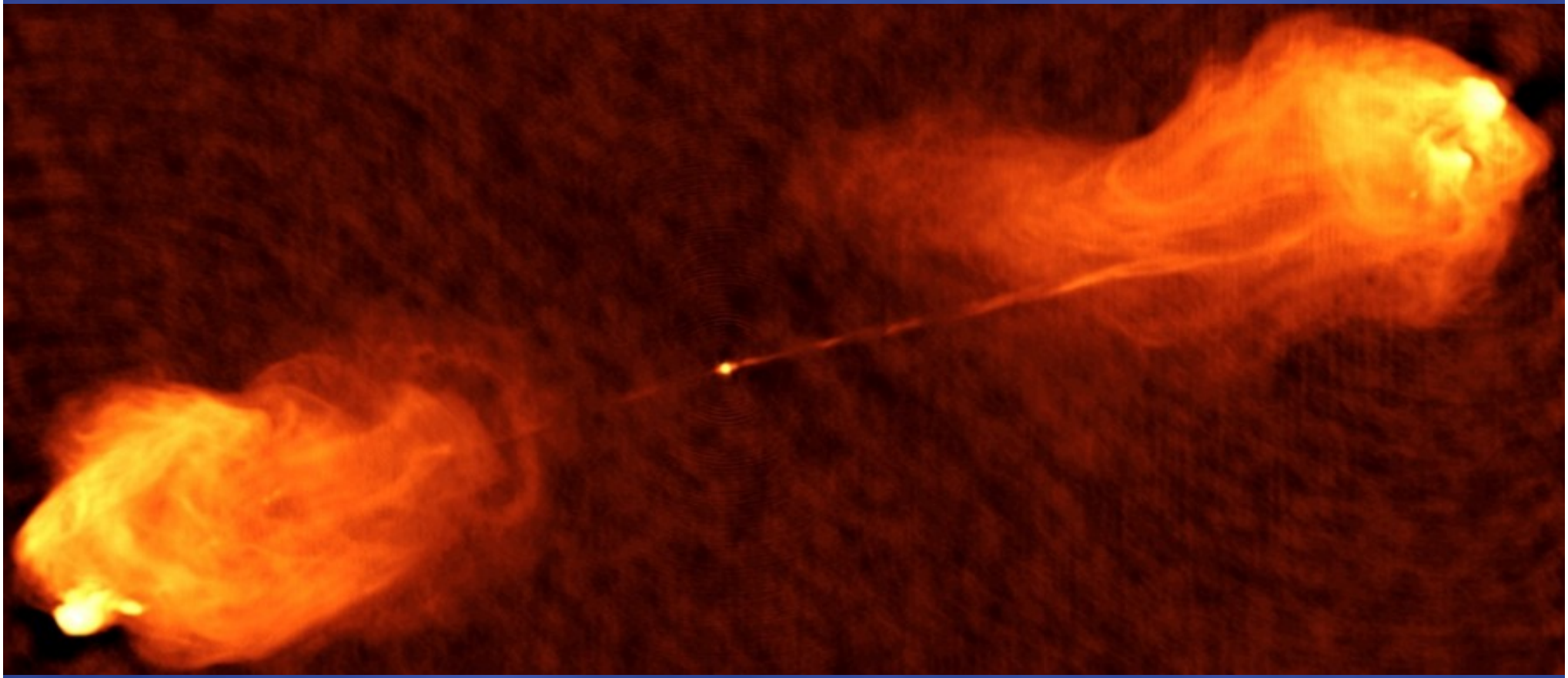
Signal Processing Tools for Radio Astronomy

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TU Delft, The Netherlands

in close collaboration with ASTRON

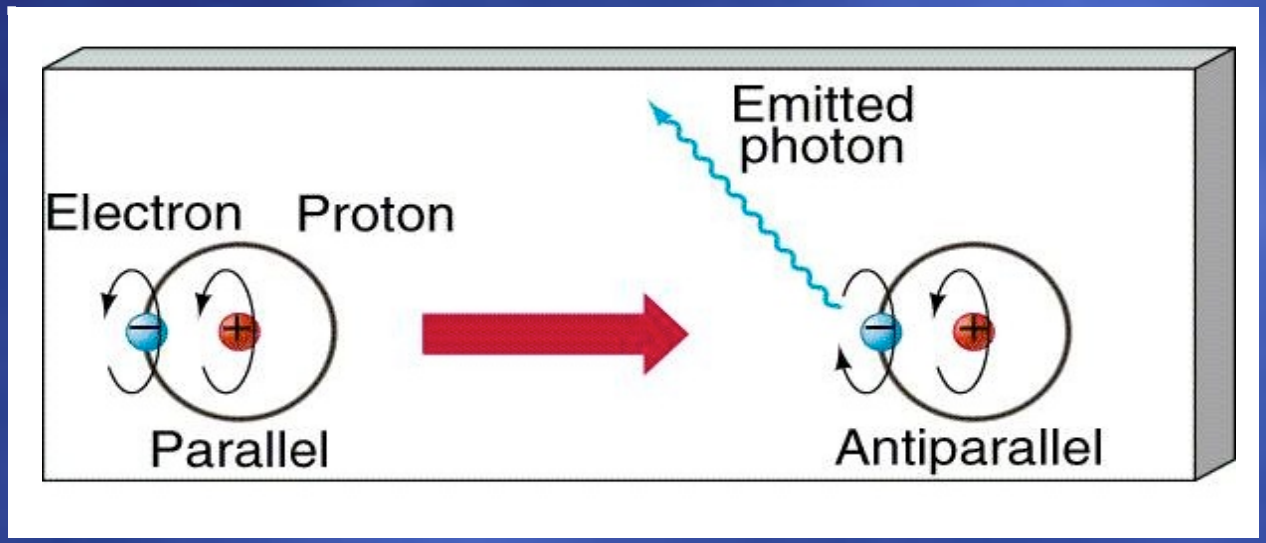
Introduction—the science

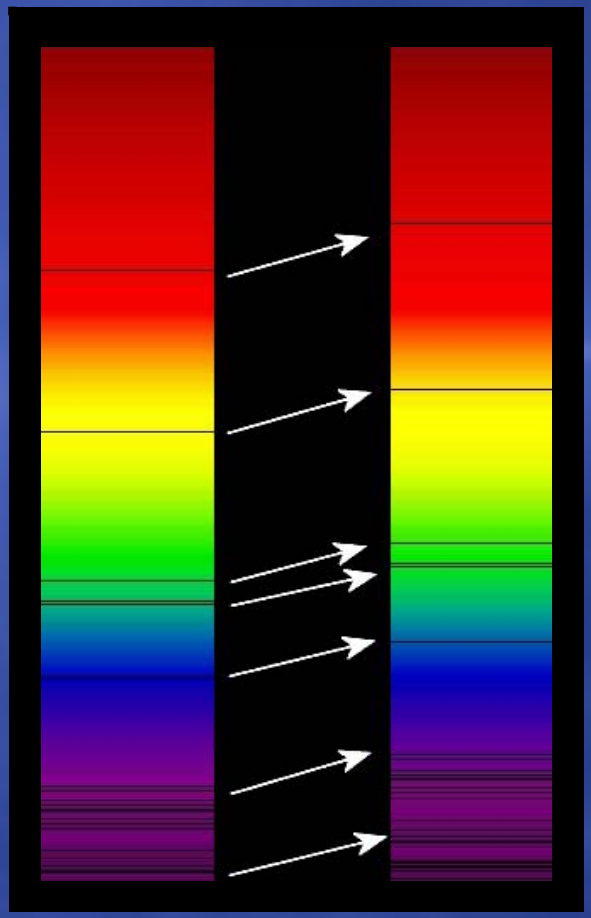
- Cygnus A (a quasar)



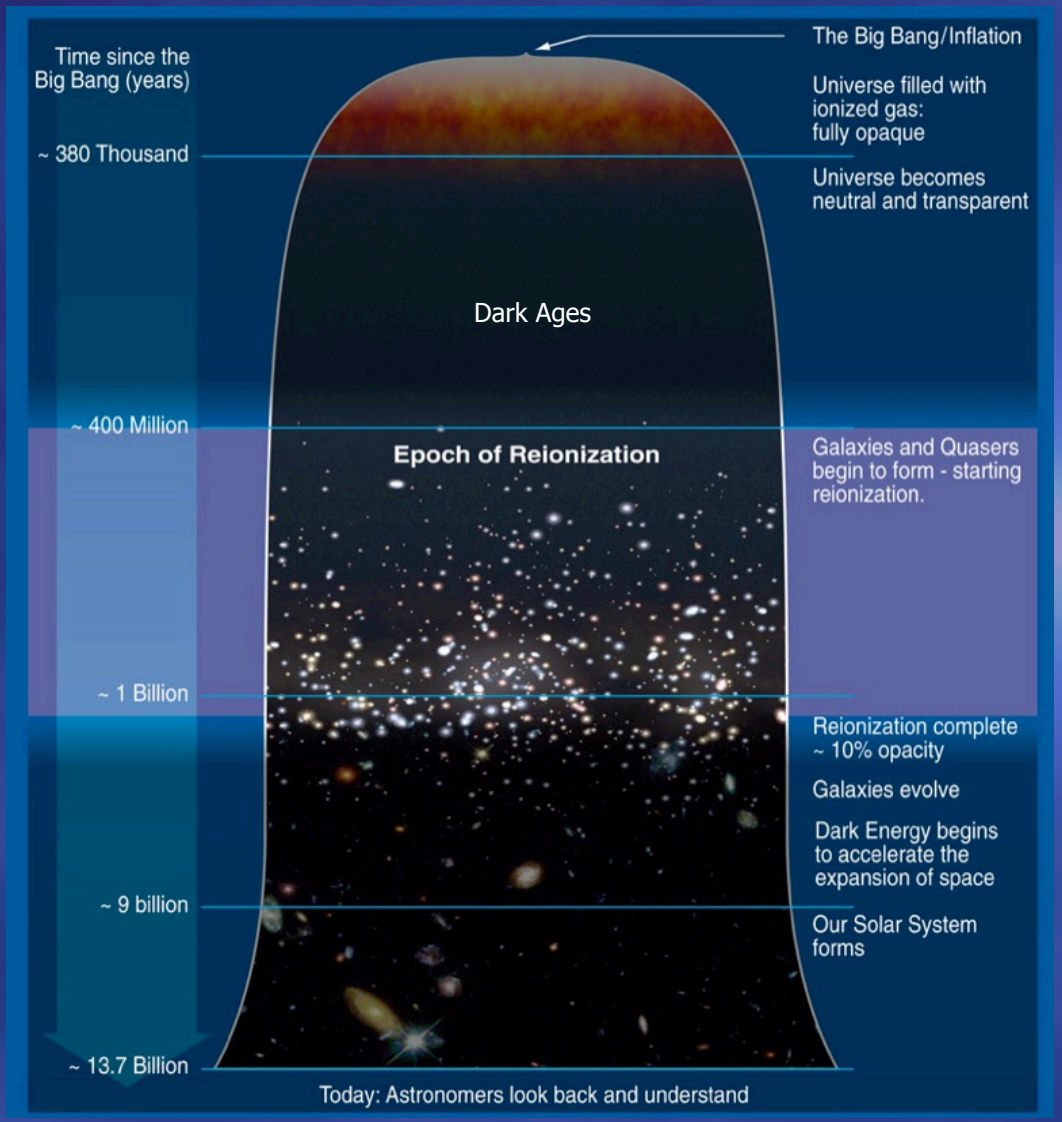
Introduction—the science

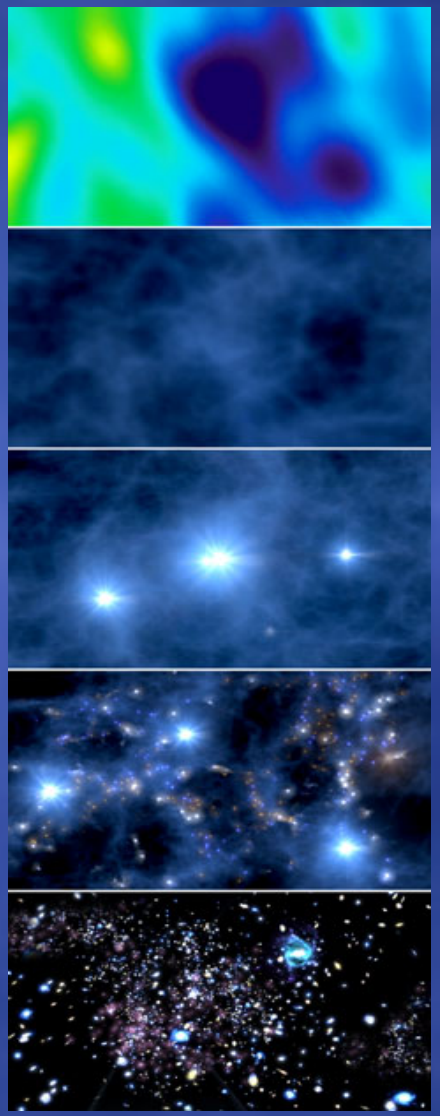
- Spectral line due to neutral hydrogen (1420 MHz)
- Redshift (“doppler”)--up to a factor 10!





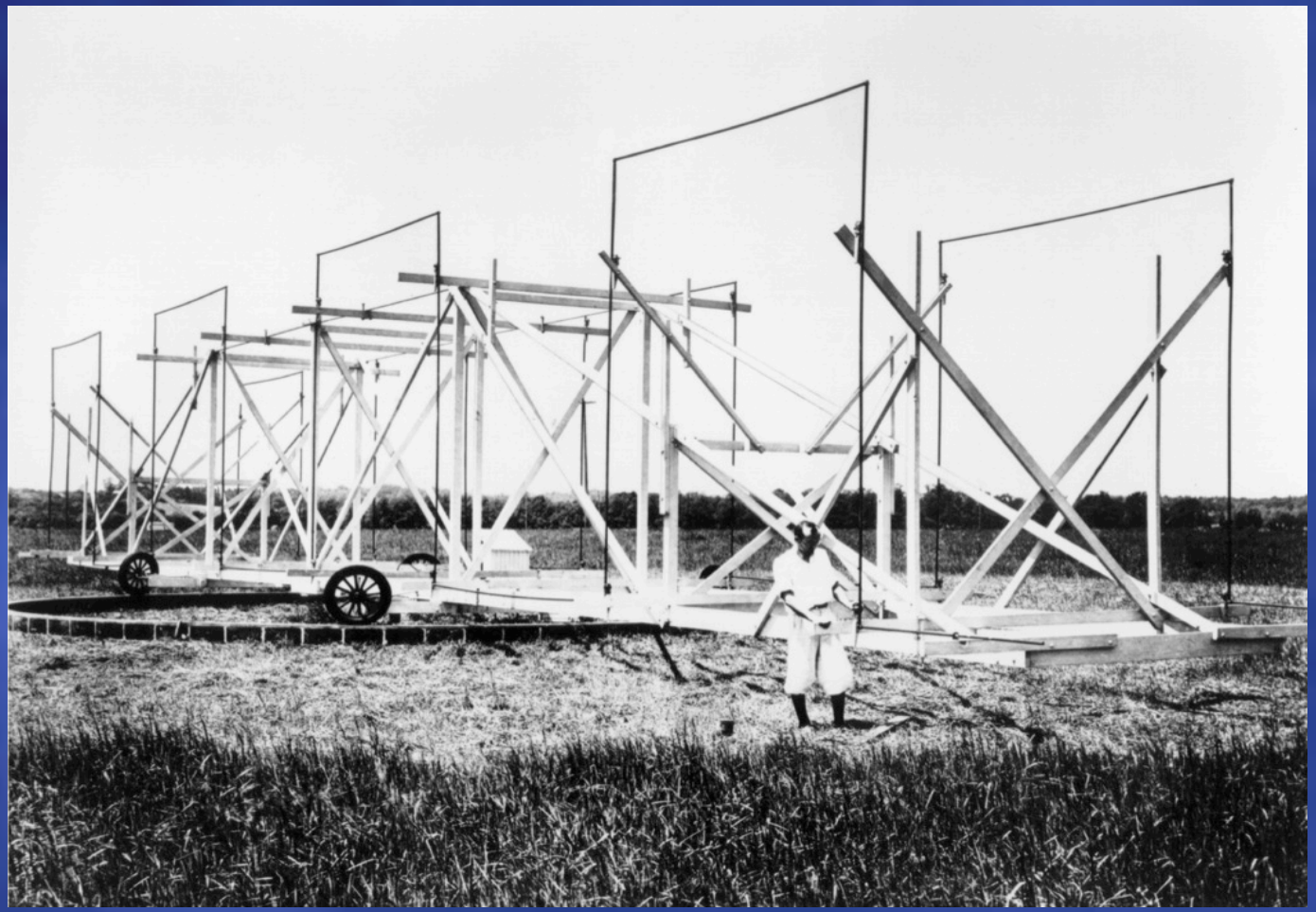
Introduction—the science





Introduction—the instruments

- Karl Jansky, Bell Labs (1928), 20 MHz



Introduction—the instruments

- Oort/van der Hulst (1950s), prediction and discovery of the HI 21 cm line (1420 MHz)



- Kootwijk (Netherlands)



- Dwingeloo (Netherlands), 1956

Introduction—the instruments

- Effelsberg, Germany (1972) 100m
- Arecibo, Puerto Rico (1960) 305m





Introduction—the instruments

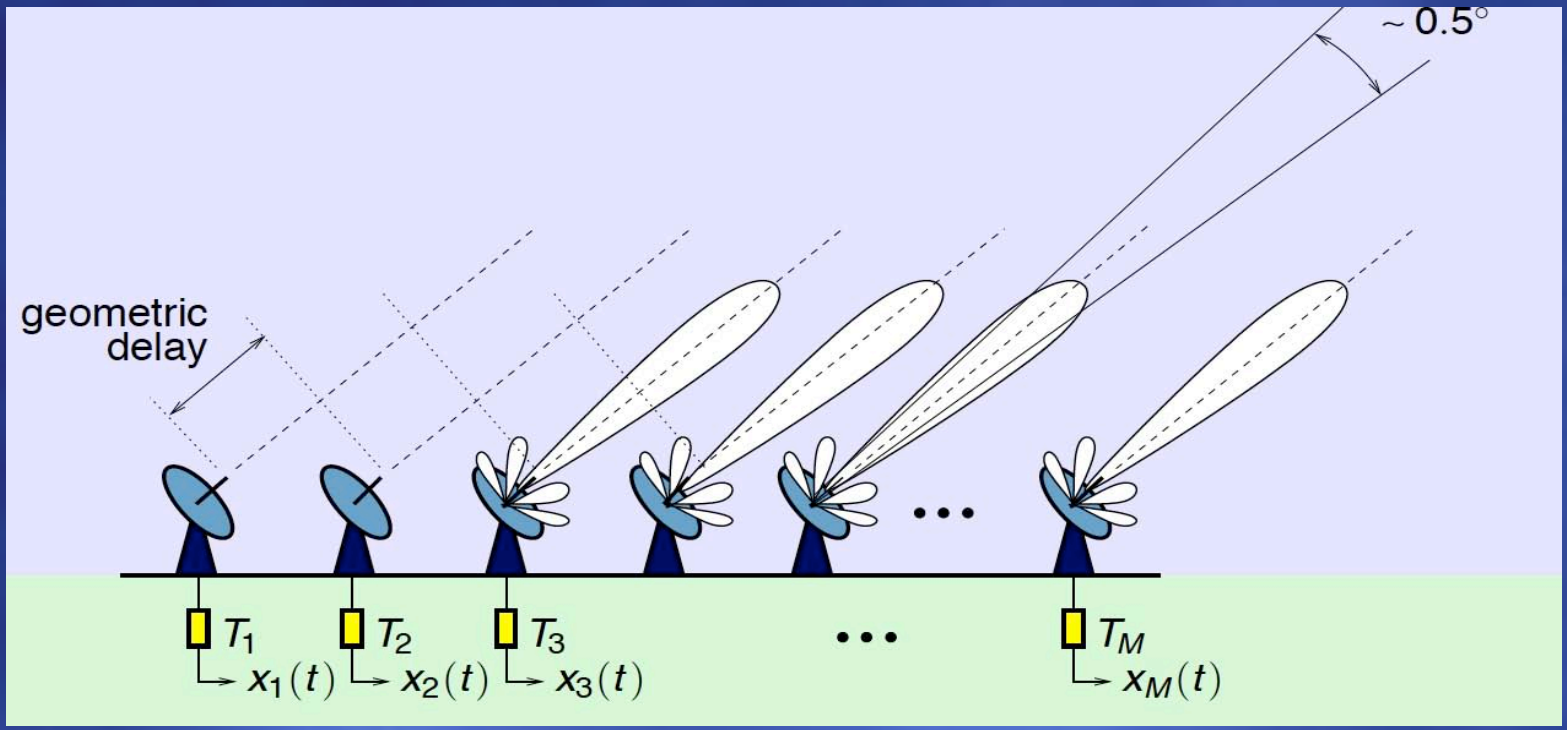
- Westerbork, Netherlands (1970), 14 dishes



- Very Large Array, New Mexico (1980), 27 dishes



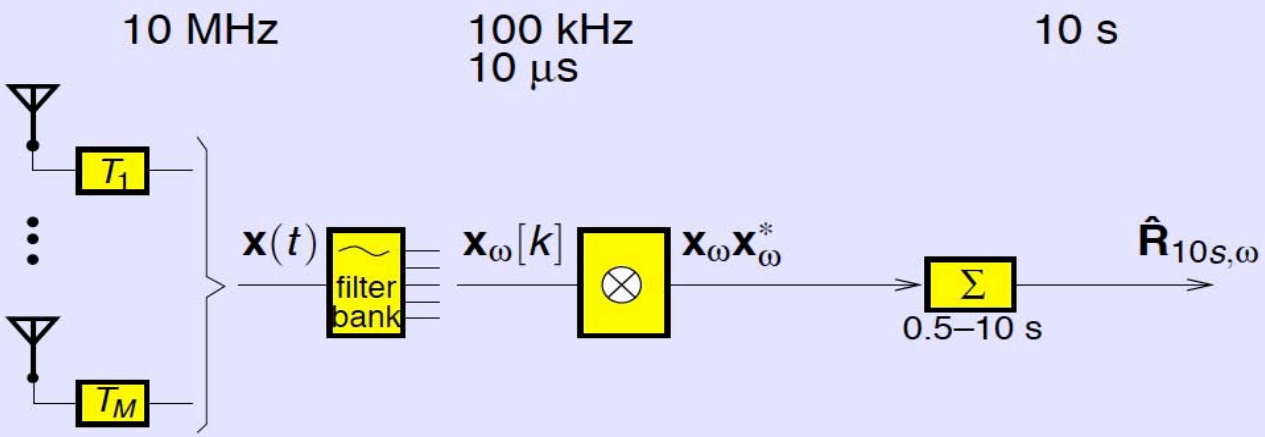
The correlation process



- Signals are stacked in vectors: $\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_M(t) \end{bmatrix}$
- The location of antenna i at time t is $\mathbf{r}_i(t)$.

  The vector $\mathbf{r}_i(t) - \mathbf{r}_j(t)$ between antennas i and j is a *baseline*.

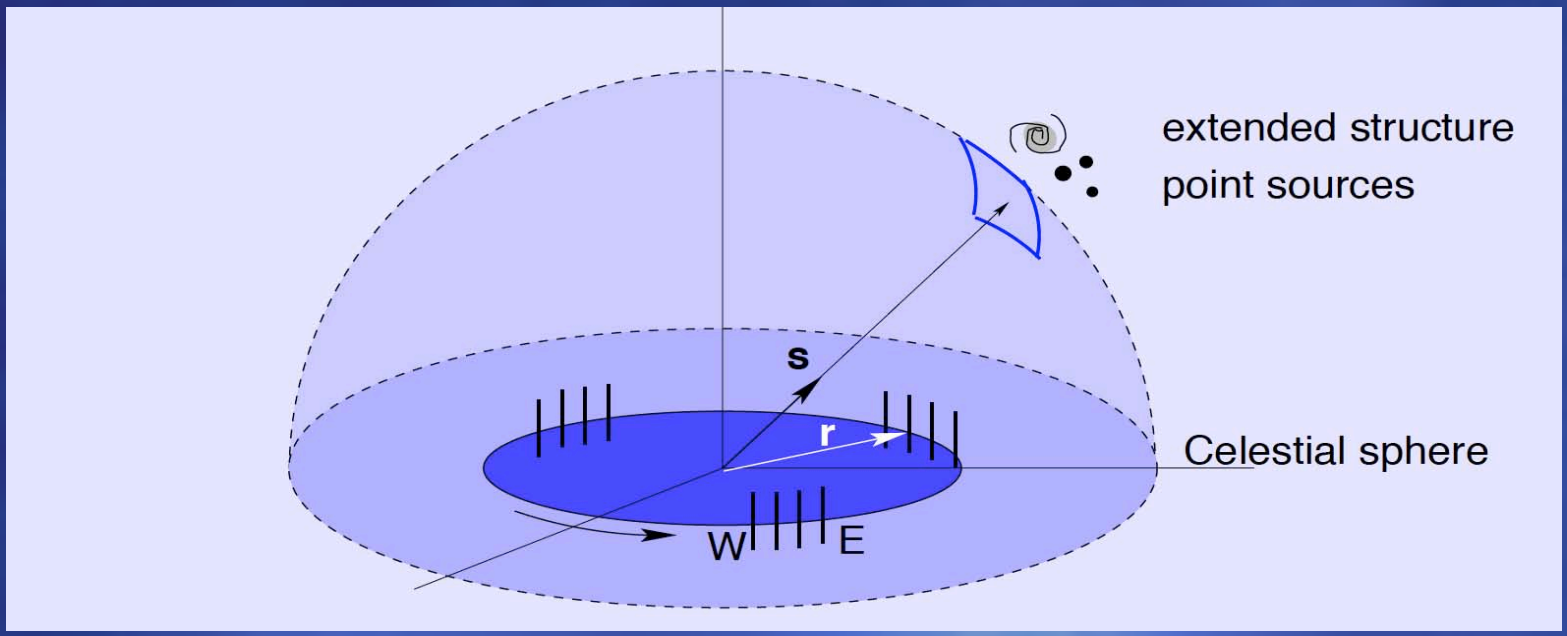
The correlation process



$$\mathbf{R}_{k,\omega} = \mathrm{E}\{\mathbf{x}_\omega(t_k)\mathbf{x}_\omega(t_k)^{\mathrm{H}}\} \simeq \frac{1}{N}\sum_{nT=0}^{10s}\mathbf{x}_\omega(t_k+nT)\mathbf{x}_\omega(t_k+nT)^{\mathrm{H}}$$

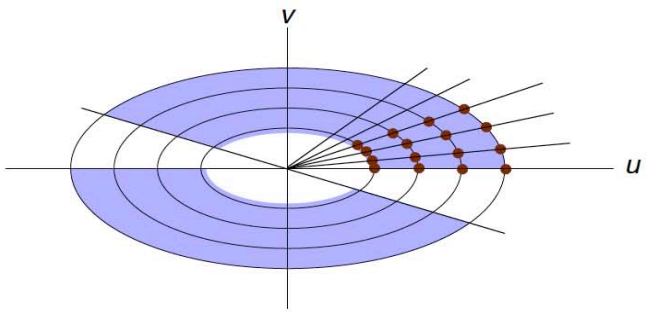
- Entry (i,j) of $\mathbf{R}_{k,\omega}$ corresponds to a *visibility* $V[\mathbf{r}_i(t_k),\mathbf{r}_j(t_k)]$ at frequency ω.

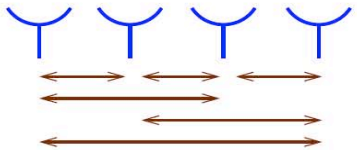
The correlation process



- We can measure correlations only for a discrete set of baselines $\mathbf{r}_i(t_k) - \mathbf{r}_j(t_k)$:

  (they are represented in normalized (u, v)-*coordinates*)

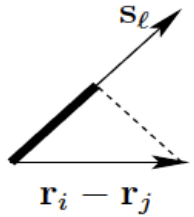




The correlation process

Simplified measurement equation

$$V_{ijk} \equiv V[\mathbf{r}_i(t_k), \mathbf{r}_j(t_k)] \quad = \quad \sum_{\ell=1}^{d} I(\mathbf{s}_\ell)\, e^{-j\,\mathbf{s}_\ell^T(\mathbf{r}_i - \mathbf{r}_j)}$$



- $I(\cdot)$ is the brightness image ('map') of interest
- $\mathbf{s}_\ell$ is the unit direction vector of the ℓ-th source (assuming discrete source model)
- V_{ijk} is the measured correlation $(\mathbf{R}_k)_{ij}$ between antennas i and j at time t_k

Classical Fourier-based imaging

Given many samples of V_{ijk}, we can compute $I(\mathbf{s}_\ell)$ via an inverse Fourier transform:

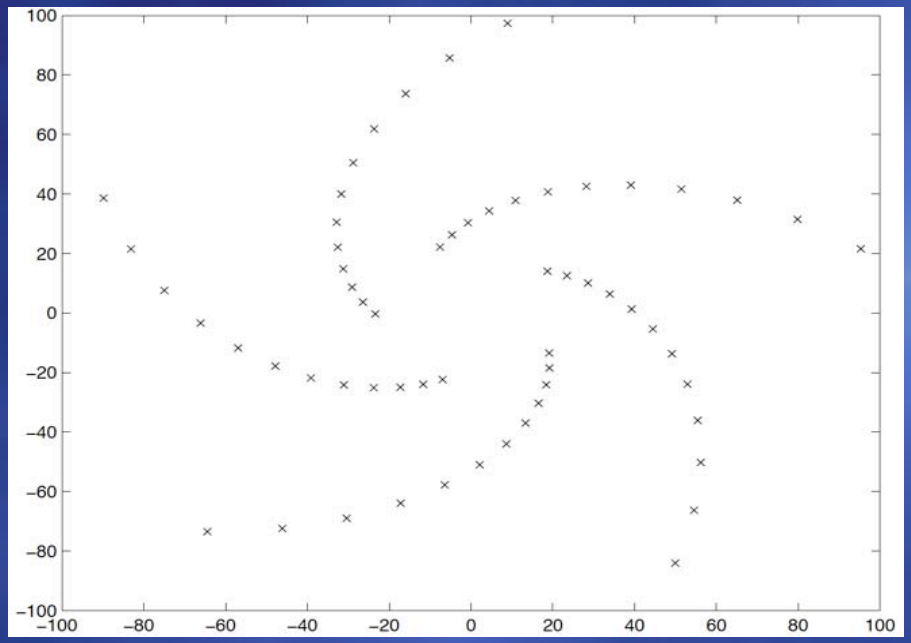
"dirty image" : $$I_D(\mathbf{s}) := \sum_{i,j,k} V_{ijk}\, e^{j\mathbf{s}^T(\mathbf{r}_i - \mathbf{r}_j)} =: \sum_{\ell} I(\mathbf{s}_\ell)\, B(\mathbf{s} - \mathbf{s}_\ell) = I * B$$

"dirty beam" : $$B(\mathbf{s} - \mathbf{s}_\ell) := \sum_{i,j,k} e^{j\,(\mathbf{s}-\mathbf{s}_\ell)^T(\mathbf{r}_i - \mathbf{r}_j)}$$

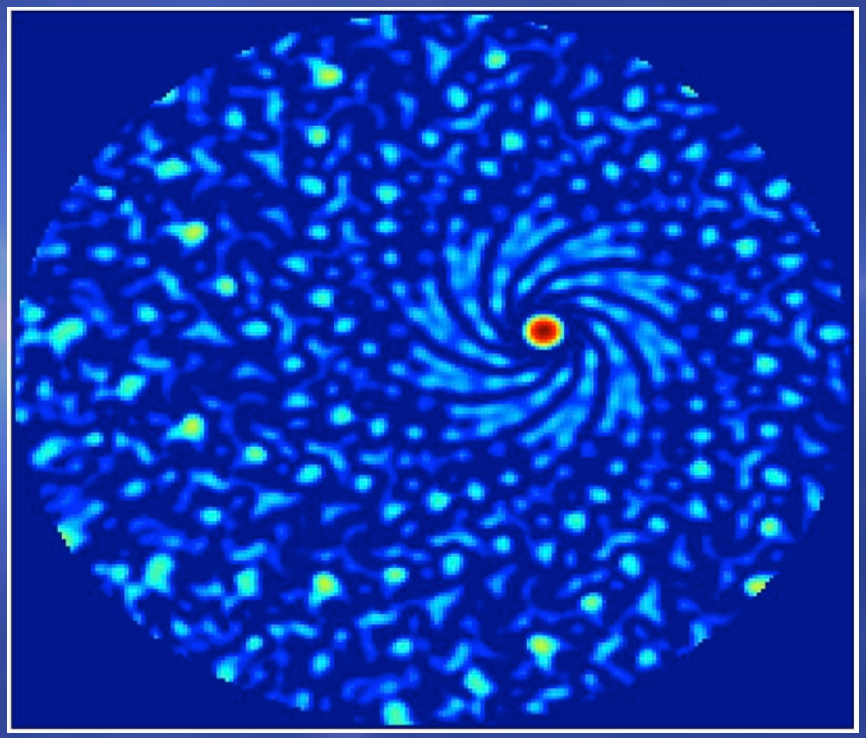
Every point source excites a beam $B(\cdot)$ centered at its location $\mathbf{s}_\ell$

The dirty image

- Example antenna configuration



- Resulting beampattern (snapshot)



The dirty image

Matrix model

- Recall

$$(\mathbf{R}_k)_{ij} \equiv V_{ijk} = \sum_{\ell=1}^{d} I(\mathbf{s}_\ell)\, e^{-j\mathbf{s}_\ell^T(\mathbf{r}_i-\mathbf{r}_j)} = \sum_{\ell=1}^{d} e^{-j\mathbf{s}_\ell^T\mathbf{r}_i}\cdot I(\mathbf{s}_\ell)\cdot e^{j\mathbf{s}_\ell^T\mathbf{r}_j}$$

- In standard array signal processing notation, we have $\mathbf{R}_k = \mathbf{A}_k \boldsymbol{\Sigma}_s \mathbf{A}_k^{\mathrm{H}}$

  where

$$\mathbf{A}_k = [\mathbf{a}_k(\mathbf{s}_1), \cdots, \mathbf{a}_k(\mathbf{s}_d)], \qquad \boldsymbol{\Sigma}_s = \begin{bmatrix} I(\mathbf{s}_1) & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & I(\mathbf{s}_d) \end{bmatrix}$$

- $\mathbf{a}_k(\mathbf{s}) = \begin{bmatrix} e^{-j\mathbf{s}^T\mathbf{r}_1(t_k)} \\ \vdots \\ e^{-j\mathbf{s}^T\mathbf{r}_M(t_k)} \end{bmatrix}$ is the array response vector.

$$I_D(\mathbf{s}) = \sum_{i,j,k} V_{ijk} e^{j\mathbf{s}^T\mathbf{r}_i}\cdot e^{-j\mathbf{s}^T\mathbf{r}_j} = \sum_{i,j,k} (\mathbf{R}_k)_{ij}(\bar{\mathbf{a}}_k(\mathbf{s}))_i(\mathbf{a}_k(\mathbf{s}))_j = \sum_k \mathbf{a}_k^{\mathrm{H}}(\mathbf{s})\mathbf{R}_k\mathbf{a}_k(\mathbf{s})$$

Imaging via adaptive beamforming

- Imaging via classical beamforming

$$I_D(\mathbf{s}) = \sum_k \mathbf{a}_k^H(\mathbf{s})\mathbf{R}_k\mathbf{a}_k(\mathbf{s})$$

- Minimum variance (MVDR) beamforming
  - Pseudo-spectrum: $I'_D(\mathbf{s}) := \sum_k \mathbf{w}_k^H(\mathbf{s})\mathbf{R}_k\mathbf{w}_k(\mathbf{s})$
  - MVDR criterium:

$$\hat{\mathbf{w}}_k(\mathbf{s}) = \arg\min_{\mathbf{w}_k} \mathbf{w}_k^H\hat{\mathbf{R}}_k\mathbf{w}_k \qquad \text{such that} \qquad \mathbf{w}_k^H\mathbf{a}_k(\mathbf{s}) = 1$$

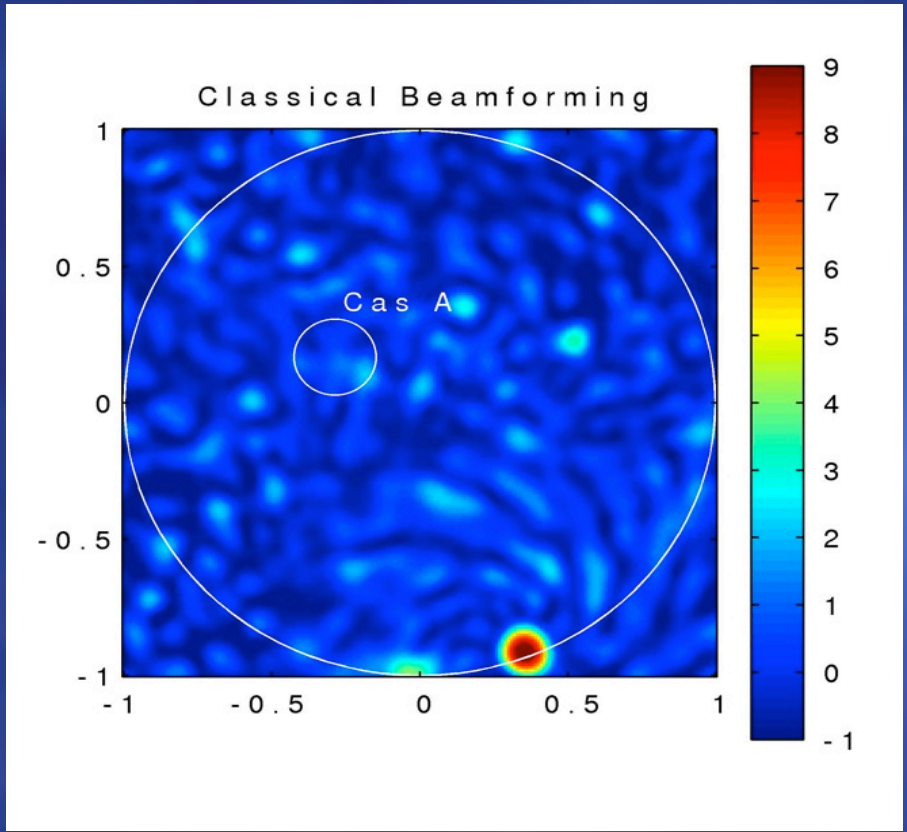
  - The solution to this problem is

$$\hat{\mathbf{w}}_k = \beta_k\hat{\mathbf{R}}_k^{-1}\mathbf{a}_k(\mathbf{s}), \qquad \text{where} \qquad \beta_k = \frac{1}{\mathbf{a}_k^H(\mathbf{s})\hat{\mathbf{R}}_k^{-1}\mathbf{a}_k(\mathbf{s})}$$

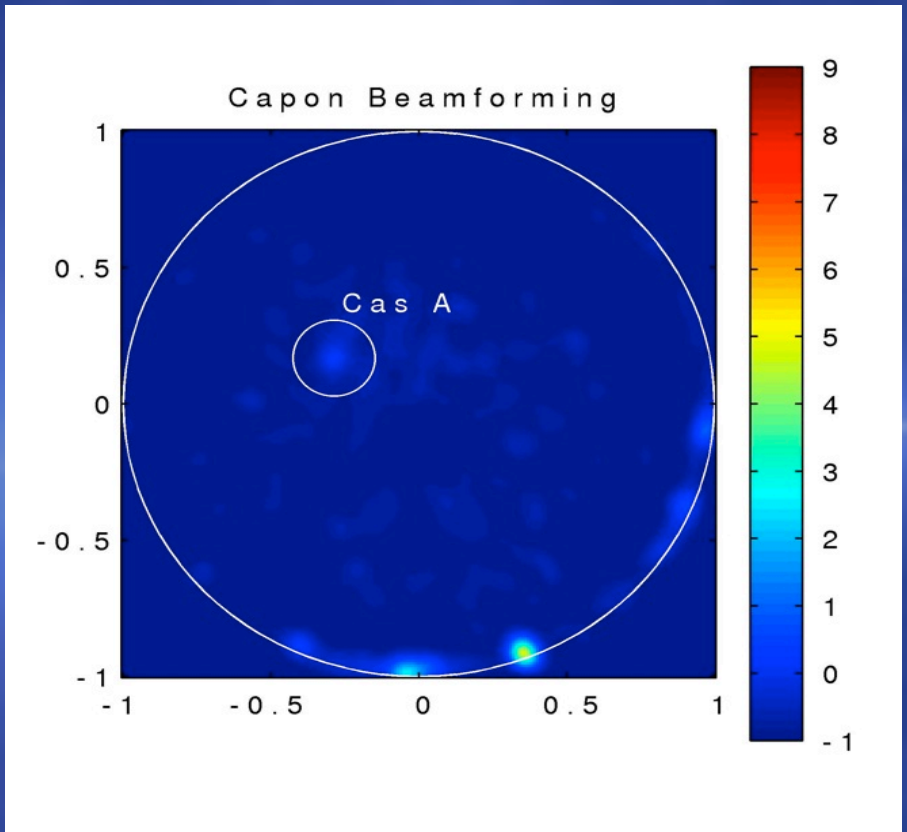
$$\Rightarrow \qquad I'_D(\mathbf{s}) = \sum_{k=1}^{K} \frac{1}{\mathbf{a}_k^H(\mathbf{s})\hat{\mathbf{R}}_k^{-1}\mathbf{a}_k(\mathbf{s})}$$

Dirty images from LOFAR teststation

- Classical beamforming



- MVDR



Imaging – matrix formulation

- Data model:

$$\mathbf{R}_k = \mathbf{A}_k \boldsymbol{\Sigma}_s \mathbf{A}_k^H + \boldsymbol{\Sigma}_n \,; \qquad \hat{\mathbf{R}}_k = \mathbf{R}_k + \text{noise}$$

- Vectorized data model: $\mathbf{r} = \text{vec}(\mathbf{R})$; vectorized image: $\boldsymbol{\sigma}_s = \text{vecdiag}(\boldsymbol{\Sigma}_s)$

$$\mathbf{r}_k = (\bar{\mathbf{A}}_k \circ \mathbf{A}_k)\boldsymbol{\sigma}_s + \boldsymbol{\sigma}_{n,k} = \mathbf{M}_k \boldsymbol{\sigma}_s + \boldsymbol{\sigma}_{n,k} \,,$$

- Stack for all k: $\mathbf{r} = \mathbf{M}\boldsymbol{\sigma}_s + \boldsymbol{\sigma}_n$

- *Dirty image:* $\hat{\boldsymbol{\sigma}}_d = \mathbf{M}^H \hat{\mathbf{r}} \approx (\mathbf{M}^H \mathbf{M})\boldsymbol{\sigma}_s$

- *Least-squares image:* $\hat{\boldsymbol{\sigma}}_s = (\mathbf{M}^H \mathbf{M})^{-1} \mathbf{M}^H \hat{\mathbf{r}} = (\mathbf{M}^H \mathbf{M})^{-1} \hat{\boldsymbol{\sigma}}_d \approx \boldsymbol{\sigma}_s$

  Can first project out $\boldsymbol{\sigma}_n$ and include weighting with noise covariance matrix

  Can be ill-posed if number of pixels in image is large

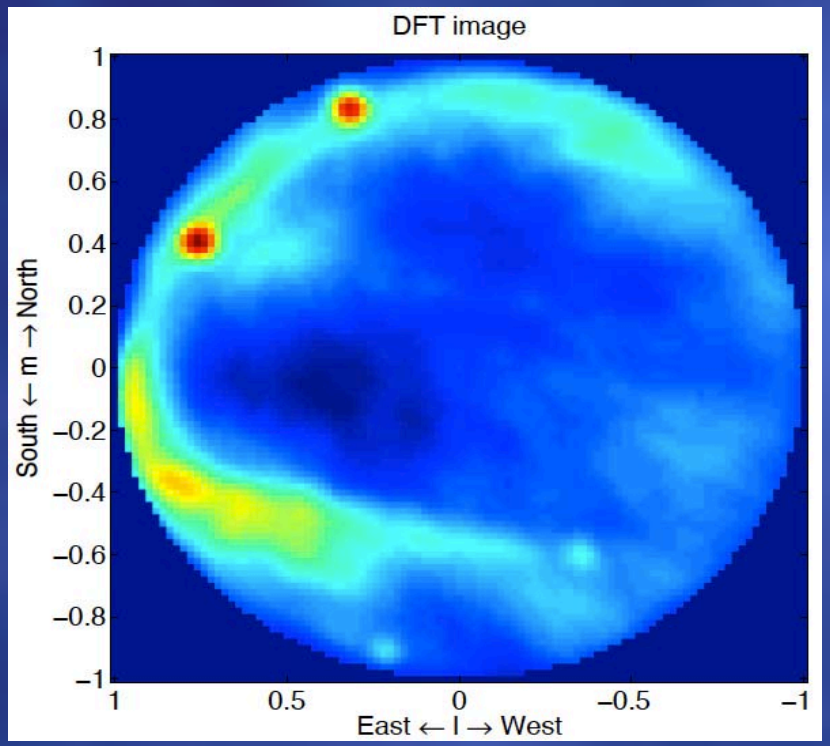
- *Compressive sampling formulation:* e.g. (Wiaux'09)

$$\min_{\boldsymbol{\sigma}} \|\hat{\mathbf{r}} - \mathbf{M}\boldsymbol{\sigma}\|_2^2 + \lambda \|\boldsymbol{\sigma}\|_1$$

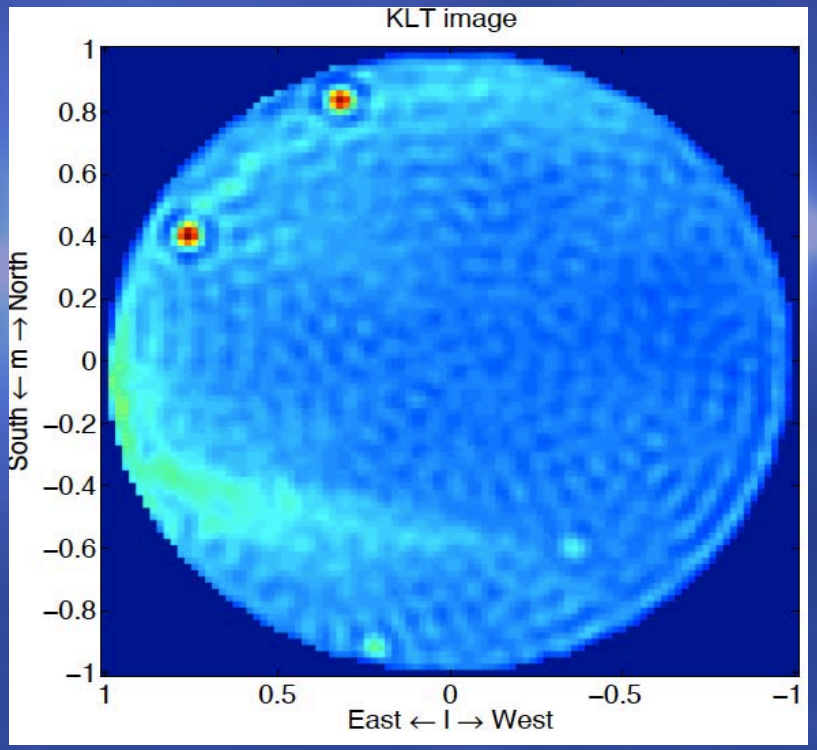
Imaging – matrix formulation

First test results of WLS imaging on LOFAR station data

- Classical dirty image (DFT)



- Weighted LS with truncation



LOFAR radio telescope array

Design parameters

- 40 stations (Netherlands)
- 8 additional stations in Germany, UK, Sweden, France, later also in Poland, Italy, …

- ~1000 antenna elements/station
- Dense core, 24 stations within 3.5 km

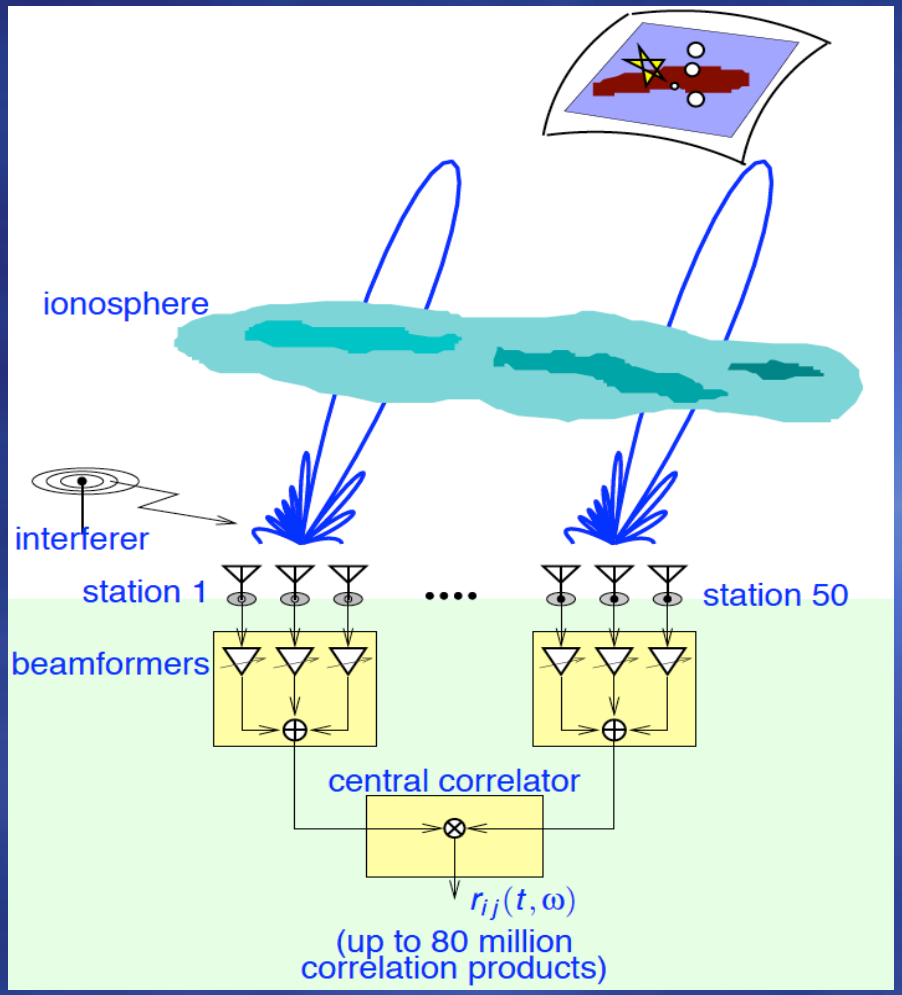
- Lowband array: 10-80 MHz
- Highband array: 120-240 MHz
- Resolution ~1 kHz

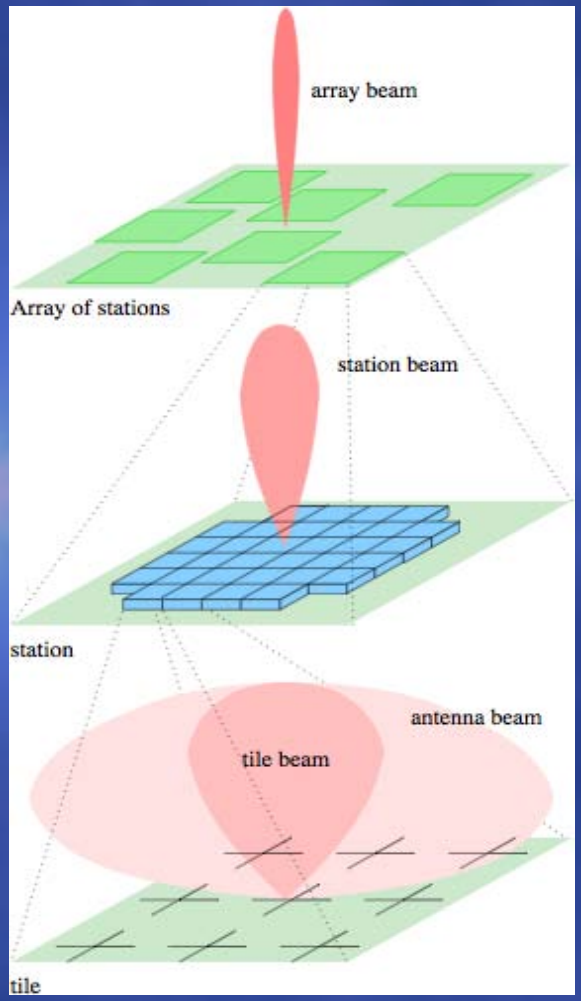
- 8 simultaneous beams
- Up to 48 MHz per beam

Designed and operated by ASTRON



LOFAR radio telescope array





LOFAR radio telescope array

Station parameters

- Each station has 2 high band array substations, and 96 low band antennas
- Beamformed output mimicks a telescope dish

- 24 core stations
- 16 NL remote stations
- 8 international stations



LOFAR core stations

Core station parameters

- `Superterp': 6 stations, diameter 300m
- 24 core stations within 3.5 km

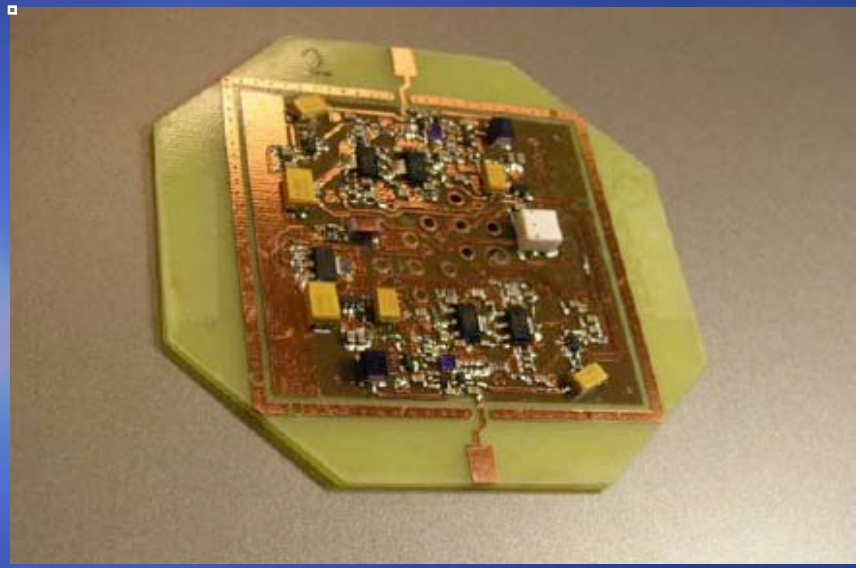




LOFAR lowband antennas

- 10-80 MHz
- Dipole antenna, 96 dipoles per station





LOFAR high-band array

- 120-240 MHz
- Each tile is 5 x 5 meter and has 4 x 4 ‘spider’ antennas encapsulated in polystyrene
- 24 tiles form a sub-station
- 2 substations per station (total 768 antennas)

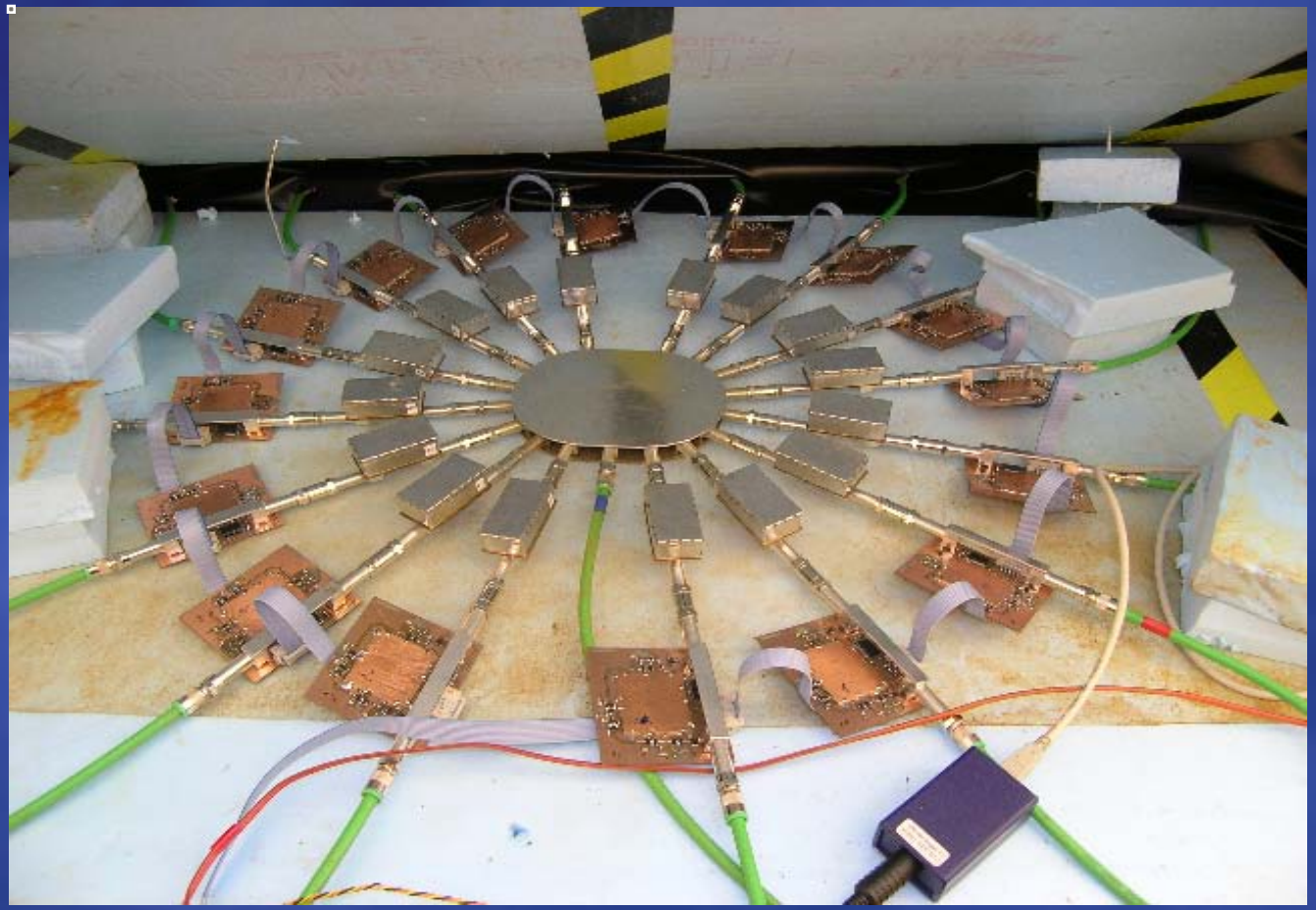






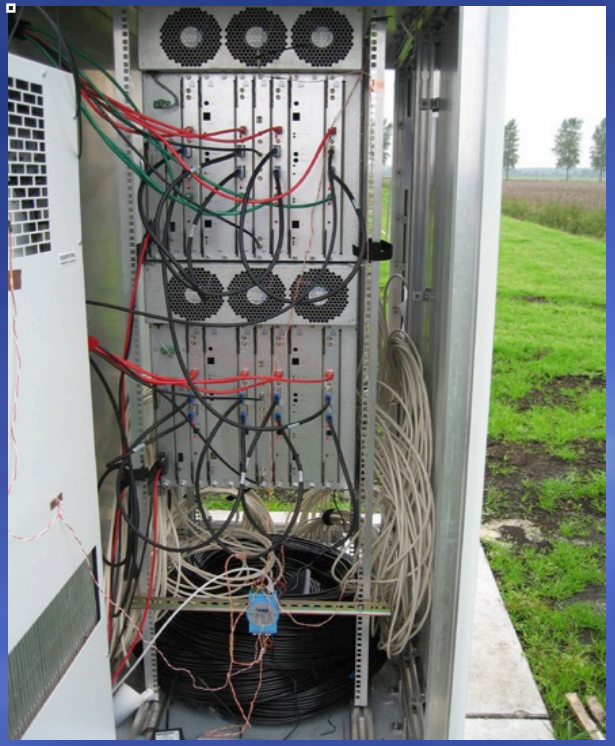
LOFAR high-band array

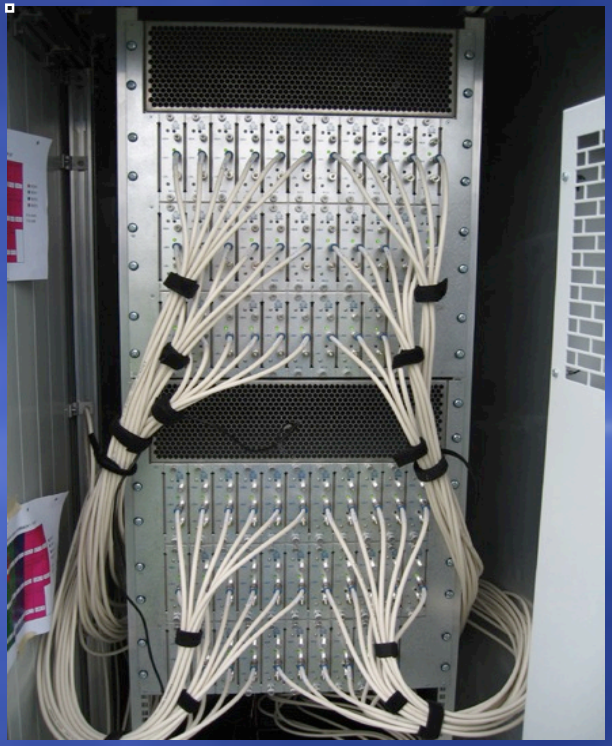
- Beamformer subsystem for a 4 x 4 tile
- 16 channels, switchable delays (5 bits)



LOFAR station hardware







LOFAR central processor

- IBM BlueGene/L
- Online: 42 Tflops peak
- Data is buffered for 10 s, correlated, then stored for later offline processing (2 PetaB) on a Linux cluster (20 Tflops)



